

Math 142 Group Project

Purpose of the Project This project is designed to mirror the spirit of a COMAP Modeling Contest solution. The goal is not to find a single “correct” answer, but to demonstrate the modeling process: translating a real-world problem into mathematics, making and justifying assumptions, analyzing the resulting model, interpreting results in context, and critically evaluating limitations.

While you may draw on mathematical tools from other courses, your project should clearly demonstrate modeling skills developed in this class, including assumption-driven model construction, qualitative and quantitative analysis, and interpretation of results.

Directions Together, you and your group will select one of the modeling problems on BruinLearn. You will collaborate with your group to design a model (or models) which address the needs of your chosen problem. Sign up for the project at this link
<https://docs.google.com/spreadsheets/d/1zr8BNPS1wVuw7ZPD5pIbW7FvJgMs6sMdaK7anrhqdo/edit?usp=sharing>

The main components of the group project include:

1. A formal write-up of your solution in the range of 10-20 pages to the chosen modeling problem. The 10–20 page range is a guideline for the substantive modeling and analysis. Lengthy appendices for raw data or code do not count toward this range and should be included only if they support the primary text.
2. A presentation (of length no more than 20 minutes) that will be delivered to the class.
3. Evaluations of group members and of the other presentations.
4. Two consultation meetings with Professor and/or TA prior to finals day.

You can find more information regarding these assignments below.

The Write-Up This is the paper describing your solution to the modeling problem. (Do not follow the contest write-up instructions.) It should include the following sections.

- It should begin with a one page summary sheet. This sheet should concisely describe the contents of the paper and should leave any reader with the impression that reading the paper will be worth their time. When writing the summary sheet, imagine a reader who will decide whether to continue reading your paper based solely on the summary sheet. (This scenario is quite common.) Although this is the first part of your paper, the summary sheet should be written last.

The following statement from the CMC is worth noting: “Summaries that are mere restatements of the contest problem, or are a cut-and-paste boilerplate from the Introduction are generally considered to be weak.”

- An introductory section which accomplishes all of the following things:
 - Gives necessary (but concise) background on the problem.

- Describes the problem (re)statement, including clarifying any ambiguous/vague language from the original problem statement.
- Defines key terminology and jargon, identifies all the relevant variables, and clarifies any non-standard or new notation that will be used.
- Identifies and justifies all the assumptions that will be made.
- Outlines the modeling approach that will be used. Provides justification of why the modeling approach is reasonable.
- The main body of the paper should include all of the following:
 - A description of the model and how it was derived, including summaries of any needed derivations and/or computations. Include illustrative examples where appropriate. Explain how the assumptions made in introductory section come into play in the model.
 - The results of applying the model to the problem, including analysis of how model outputs depend on key parameters, assumptions, or inputs. This may include (but is not limited to) parameter exploration (fixed points), limiting cases (long term behavior), comparisons across scenarios, or qualitative reasoning about model behavior.
 - A model assessment which critically addresses the strengths and weaknesses of the model in relation to the original problem goals. This section should explicitly discuss the impact of assumptions, identify situations in which the model is likely to fail, and, if possible, describe how the model could be tested, calibrated, or improved with additional data or revised assumptions.
 - At some point, your group should consider at least one alternative modeling choice or refinement (e.g., a different assumption, functional form, or level of complexity) and briefly discuss how it would change the results or interpretation.
- A conclusion section summarizing results and findings. Should also discuss possible future directions, e.g., discuss how the current model could be strengthened or discuss possible other applications of your modeling approach.
- A reference section documenting all the resources that were used including report on all AI use. Refer to the AI use policy. This section does not count towards your page count.

You may use mathematical tools or techniques from other courses (e.g., linear algebra, probability, differential equations, optimization). However, any such tools must be clearly explained, justified, and interpreted within the context of the model. Unexplained “black-box” methods will receive limited credit.

The Presentation Your group is required to give a presentation to the class summarizing your modeling approach and your findings. The presentation should clearly explain what was learned from the model, not just what was done. The presentation must include slides and every member of the group must speak for a roughly equal amount of time during the presentation. The presentation must not exceed 20 minutes plus 5 minutes for questions.

Peer & Presentation Evaluations: While classmate feedback on presentations does not affect your group’s technical score, the submission of these evaluations is a required component of your grade. Failure to submit thoughtful evaluations for your peers will result in a loss of these points.

- *Group Member Evaluation:* The workload of the group should be equitably divided between members of the group. Each member will evaluate the rest of the group and submit the evaluations. These will be used for situations in which a member does not contribute an equal amount.
- *Presentation Day Evaluation:* Everyone should be giving each group their attention. Short evaluations for each presentation will be completed by everyone to get feedback on the various projects. Classmate evaluations will not affect your groups score.

Consultation Meeting Logistics & Evaluation: There are exactly two mandatory progress consultations with the Professor and/or TA. These function as both a consultative resource and a cumulative integrity check.

Consultation 1: Initial Model Formulation & Proof of Work

- *The "Proof of Work" Requirement:* At the first meeting, groups must present the "ancillary" materials of their modeling process. This includes handwritten derivations, trial-and-error notes, or initial code drafts. A "perfect" model that appears without a documented history of group discussion and revision will be scrutinized.
- *Individual Accountability:* Because modeling requires judgment, every student is expected to understand the full paper. If one member is unable to explain a specific derivation or parameter choice during a meeting, the professor reserves the right to adjust individual grades, regardless of the group's technical output.

Consultation 2: The Analysis & Defense Check-in

- *The "Proof of Work" Requirement:* The second meeting focuses on the technical execution of your model. Groups must demonstrate that the mathematical analysis—including any derivations, computations, or simulations—is the result of independent group reasoning. You should be prepared to explain the "why" behind your choice of analysis techniques (e.g., sensitivity analysis, fixed points, or scenario comparisons).
- *Individual Oral Defense:* During this meeting, the Professor or TA may ask any individual group member to explain a specific part of the analysis. Inability to explain how a result was derived or what a specific parameter represents will be taken as evidence of inappropriate AI use.

Consistent with the AI Policy, the goal of these meetings is to ensure you are the "pilot" of the project. If you use AI for formatting or grammar, that is permitted, but the mathematical reasoning must be yours. Bringing your "rough drafts" and "failed attempts" to these meetings is the best way to prove the final product is your own.

Grading In total, the project will be graded out of 150 points.

Integrity & Process

- 5pts Deadlines & Group Logistics – Timely sign-up, topic decision, etc
- 25pts Consultation 1: Initial Model Formulation & Proof of Work (Focus on original derivation and justification of assumptions)

- 15pts Consultation 2: Progress & Analysis Defense (Focus on mathematical execution and individual understanding)
- 5pts Evaluations Submitted – Peer evaluation and presentation feedback forms.

Presentation

- 10pts Technical Communication – Effective use of slides and clear delivery of modeling insights

Model/Paper

- 30pts Model Design – Selection of variables, logical derivation, and justification of realistic assumptions
- 25pts Mathematical Analysis – Correct application of tools (qualitative and quantitative) and exploration of model behavior
- 15pts Interpretation & Insight – Connecting results back to the real-world problem and evaluating limitations
- 15pts Technical Documentation – Flow, grammar, and incorporation of all figures and tables. Figures and tables need to be labeled and explained. All sections included as well.
- 5pts Citation/References – Background research performed and incorporated

Disclaimer: In the event that workload is not distributed equally among group members, the professor may assign individual members a grade that is different than the rest of the group.

AI Policy Instructions

You are still responsible for reading COMAP's AI Policy in its entirety.

Permitted Uses of AI

Students may use AI tools (e.g., ChatGPT, Copilot) only for limited support tasks, including:

- Editing for grammar, clarity, or organization
- Formatting assistance (LaTeX, Word, figures)
- Generating examples of code syntax, provided the code is understood, verified, and adapted by the group for formatting graphs, labeling axes, or creating a specific plot type (e.g., "Write Python code to plot these three specific equations with these parameter values").
- Summarizing background information or definitions, with independent verification

If AI is used to generate visualization code, students are responsible for the mathematical accuracy of the output. In consultations, you may be asked to explain how specific code parameters relate back to your model's variables to prove you are the 'pilot'.

Prohibited Uses of AI

The following non-exhaustive list lists uses of AI are not permitted and constitute academic misconduct in this course:

- Formulating or selecting the mathematical model or modeling strategy

- Choosing assumptions, variables, or parameters
- Designing or carrying out the analysis. You may not feed your model or equations into an AI and ask it to perform the analysis for you. This includes asking AI to find fixed points, determine stability, perform sensitivity checks, or interpret the behavior of the system.
- Interpreting results or drawing conclusions
- Writing substantive portions of the paper, summary sheet, or conclusions
- Replacing group discussion or independent reasoning with AI-generated content
- Submitting AI-generated work that you do not fully understand and cannot explain

If an AI tool meaningfully contributes to the mathematical content, modeling choices, or reasoning of the project, then it has exceeded acceptable use.

Disclosure Requirement

All AI use must be explicitly disclosed in the reference section of the paper. The disclosure must include:

- The AI tool used
- The purpose for which it was used
- A brief description of how its output was incorporated or modified

Undisclosed AI use will be treated as a violation of this policy.

Consequences of Misuse

If, in the judgment of the instructor and/or TAs, AI has been used in a manner inconsistent with this policy, then:

The maximum possible score for the affected student or group on the project will be 60%, regardless of the technical quality of the submission. More serious or repeated violations may be referred for further academic integrity review, consistent with university policy.

This determination is based on professional judgment, including but not limited to inconsistencies in explanation, inability to defend work, or evidence of AI-generated reasoning.

Group Responsibility

This is a group project, and each group member is responsible for the integrity of the entire submission.

- Every student is expected to read and understand the full paper before submission.
- By signing the contribution statement, each member affirms that they believe the work complies with the AI use policy.
- Claiming ignorance of another group member's AI misuse is not an acceptable defense.

If AI misuse is identified, individual grades may differ within a group.